Year 12 Maths - Differentiation

Similar questions - Differentiating and integrating powers

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 $\mathrm{f}(x) = 2x^{3/2} - 9x + 5$ for $x > 0$.

(a) Find $\mathrm{f}'(x)$ and $\mathrm{f}''(x)$. **[3]**

(b) Find the set of values of x for which f is a decreasing function. **[2]**

(c) Find $\int \mathrm{f}(x)\,\mathrm{d}x$. **[3]**

(d) A curve has $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \mathrm{f}(x)$ and passes through $(4,0)$. Find its equation. **[3]**

2 A curve has equation $y = \dfrac{x^2+3}{\sqrt{x}}$ for $x > 0$.

(a) Write y in the form $x^p + 3x^q$ and find $\dfrac{\mathrm{d}y}{\mathrm{d}x}$. **[3]**

(b) Show that the curve has a stationary point at $x = 1$, and use $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2}$ to determine its nature. **[4]**

(c) Find $\displaystyle\int_1^4 y\,\mathrm{d}x$. **[3]**

Question 1

(a) $f'(x) = 3x^{1/2} - 9$, $f''(x) = \frac{3}{2}x^{-1/2}$

(b) $3\sqrt{x} - 9 < 0$, so $\sqrt{x} < 3$, giving $0 < x < 9$

(c) $\frac{4}{5}x^{5/2} - \frac{9}{2}x^2 + 5x + c$

(d) $y = \frac{4}{5}x^{5/2} - \frac{9}{2}x^2 + 5x + c$. At $x = 4$: $\frac{4}{5}(32) - 72 + 20 + c = 0$, so $c = 26.4$. $y = \frac{4}{5}x^{5/2} - \frac{9}{2}x^2 + 5x + \frac{132}{5}$

Question 2

(a) $y = x^{3/2} + 3x^{-1/2}$. $\dfrac{dy}{dx} = \frac{3}{2}x^{1/2} - \frac{3}{2}x^{-3/2}$

(b) At $x = 1$: $\frac{3}{2} - \frac{3}{2} = 0$. $\dfrac{d^2y}{dx^2} = \frac{3}{4}x^{-1/2} + \frac{9}{4}x^{-5/2}$, which is $3 > 0$ at $x = 1$, so a minimum (point $(1,4)$)

(c) $\left[\frac{2}{5}x^{5/2} + 6x^{1/2}\right]_1^4 = \left(\frac{64}{5} + 12\right) - \left(\frac{2}{5} + 6\right) = \frac{62}{5} + 6 = \frac{92}{5} = 18.4$